

EduCRM System Architecture & Workflow Diagrams

End-to-End Application Lifecycle, Role-Based Access Control (RBAC), System Sequence Diagram (SSD), and Component Interaction Sequence Diagram (SD).

PLATFORM	LIVE PRODUCTION URL	SECURITY MATRIX
EduCRM Enterprise Platform	https://education-crm-9fee2.web.app	Role-Based Stage Authorization (RBAC)

1. End-to-End Role-Based State Workflow

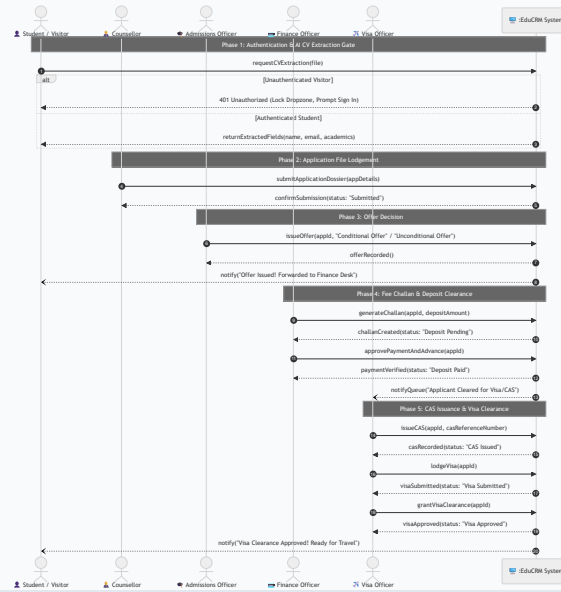
Visualizes the progression of student applications through dedicated organizational desks. Stage changes are enforced strictly via stageAuthorization.ts.

EduCRM Architecture Document

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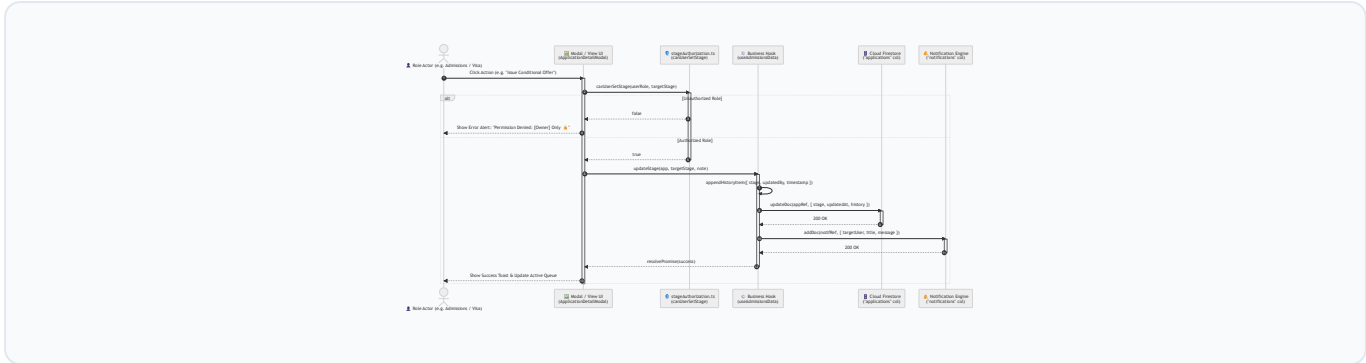
2. System Sequence Diagram (SSD) — Black-Box Behavioral View

Illustrates the external system events and responses between human role actors and the EduCRM system boundary, enforcing prerequisite conditions at each gate.



3. Detailed Sequence Diagram (SD) — Component & DB Flow

Details internal class/module collaboration across View components, the central RBAC utility, business state hooks, and Google Cloud Firestore.



4. Role-Based Stage Authorization Matrix

Application Stage	Responsible Department	Authorized User Roles	Description & Permissions
Draft / Review / Submitted	Counselling	counsellor, team_leader, office_manager	Student onboarding, document verification, application submission to university.
Conditional / Unconditional Offer	Admissions	admissions_officer, university_partner	Academic assessment, prerequisite evaluation, and issuing offer letters.
Deposit Pending / Deposit Paid	Finance	finance_officer	Generating tuition fee challan, verifying payment receipts, fee clearance.
CAS Issued / Visa Submitted / Approved	Visa	visa_officer	CAS/COE logging, embassy visa lodgement, and immigration grant clearance.
Universal Override (All Stages)	Admin	platform_super_admin, org_admin	Full operational oversight and administrative exception management across all stages.

